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Neuronal chemistry and functional organization in the primate visual system

Stewart H.C. Hendry and David J. Calkins

Beginning with the first step of visual processing and proceeding outward from that point, the neurons involved in different aspects of vision are distinct. Stated simply, neurons doing different things look different. They often display distinct morphological features and they usually express different molecules. In addition, neurons that perform a common function usually aggregate together to form recognizable layers or compartments that can be studied in isolation because they are neurochemically distinct. Here is found, then, a junction of two major domains in neuroscience research, as discovery of molecular diversity among neurons is exploited to study organization and function of the primate visual system.

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AS A LEGACY OF SPERRY'S chemoaffinity hypothesis¹, the expectation 15 years ago was that the selective expression of molecules would lead to the discovery of markers specific for neuronal types². For the nervous systems of invertebrates the discovery and use of type-specific markers have met with frequent success. Not so successful has been the application of molecular biology to an understanding of organization and function in the mammalian CNS. The problem is simple, as marker proteins or peptides specific for individual types of neurons have rarely been encountered. Yet when successful, most notably in studies of the primate visual system, the discovery of molecules unique to specific neuronal groups and the ability to see and study those groups in isolation have proven extremely powerful for addressing questions that are significant to organization and physiology.

These questions range from the role of identified neuronal types in the ability most primates possess to discriminate objects of different colors, to the basic division of labor in well-characterized regions of the visual system. Most successes have occurred for regions early in the visual pathway – the retina, lateral geniculate nucleus

(LGN) and the primary visual area (V1) – but recent findings suggest that much will be learned of higher-order areas of the visual cortex by the application of molecular biology to visual system function and organization.

Type-specific markers in primate retina

Sense and nonsense

Expectations for type-specific markers have been well met for the mammalian retina, where proteins or peptides are expressed by restricted populations of neurons (Table 1). These type-specific molecules divide themselves sharply into two groups: those that make sense and those that do not. The sensible group is the one in which the basic function of the cell type is a direct function of the molecule. In this context, the photoreceptors stand out by their expression of photopigments or opsins, because each type of cone and the rods absorb a restricted range of wavelengths as a direct result of the opsin each expresses^{3,17}. Thus the precise function of a photoreceptor type is predictable on the basis of its expression of an opsin, and mutations in opsin genes produce predictable outcomes in spectral sensitivity and color discrimination^{4,18}.

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TABLE 1. Immunocytochemical identification of some cell types in monkey and human retina

Cell class	Antigen	Cell type	Refs
Cone	S-cone opsin	S cone	3,4
	L- and M-cone opsins	L and M cones	3,4
Horizontal	Parvalbumin	H1 and H2	5
	Glutamate decarboxylase (65 kDa)	H1 and H2	6
Bipolar	Recoverin	Midget OFF	7
	Cholecystokinin	S-cone ON	8
	Protein kinase C (α subunit)	Rod	9
		Diffuse ON 4	10
	Glutamate transporter protein (GLT-1)	Diffuse OFF 2 and midget OFF	10
Amacrine	Calbindin	Diffuse OFF 3	10
	Calretinin	A-II	11
	Choline acetyltransferase	Starburst	12,13
	Tyrosine hydroxylase	Dopaminergic	14,15
Ganglion	GABA _A receptor	Parasol ON and OFF (fovea)	16

Type-specific expression that makes sense is also apparent at the synapses between photoreceptors and bipolar cells. All vertebrate photoreceptors hyperpolarize (turn OFF) when light is absorbed by opsins in their outer segments^{19,20}, yet bipolar cells include not only hyperpolarizing (OFF) types, in which the sign of the signal from photoreceptors is preserved, but also depolarizing (ON) types, in which the sign is reversed²¹. This dichotomy exists even though all photoreceptors release the same neurotransmitter, glutamate, at each of their synapses²². That a single photoreceptor imposes, and a single neurotransmitter induces, both sign-reversing and sign-conserving responses is a direct function of the glutamate receptor types expressed by depolarizing and hyperpolarizing bipolar cells. For example, ON bipolar cells are sensitive to the glutamate analog ($\pm$)-2-amino-4-phosphonobutyric acid (AP-4), application of which produces an outward current resulting from closure of nonspecific cation channels^{23,24}. In contrast, AP-4 has no effect on OFF bipolar cells, whereas a second glutamate analog, kainate, produces a strong inward current attributable to the opening of nonspecific cation channels²⁵. The molecular correlate of these findings is the expression by some OFF bipolar cells of the ionotropic Glu₅–Glu₇ receptors and by ON bipolar cells of the metabotropic mGlu₆ receptor²⁶.

Molecular diversity of bipolar-cell types

In contrast to expression that produces function, many type-specific proteins or peptides make no apparent sense for the function of a neuron. Three examples include bipolar-cell populations that relay signals from rods, from cones that are sensitive to short (S) wavelengths and from cones sensitive to middle (M) or long (L) wavelengths. Expression of protein kinase C (PKC)⁹, cholecystokinin (CCK)⁸ and recoverin⁷ is unique to each of these three types. Whereas much is known of the role each molecule plays, nothing can be said about what any of them contributes to the unique functional properties of the relevant bipolar cell type.

To take one example, CCK is known to be neuroactive in the retina, but what the peptide contributes to the physiology of S-cone ON bipolar cells is a mystery. Nonetheless, the pattern of CCK expression has been used to determine the overall density and distribution of these bipolar cells, their morphological variation across the retina, the level of convergence and divergence between them and the S cones, and the precise

synaptic organization of this cell type and of the neurons with which they communicate⁸. Much the same can be said for other retinal neurons whose neurochemical signatures have been discovered. Discovery of a molecule in a specific neuron type may be fortuitous, but the result is a better understanding of the role that type plays in vision.

There is a subtext to Fig. 1 and Table 1: neurochemically distinct bipolar cells carrying different types of photoreceptor signals are morphologically distinct. The difference in structure among neurons along S-cone pathways compared with M/L-cone pathways continues into the ganglion cell layer. Ganglion cells displaying both spatial and spectral antagonism for M- and L-cone inputs correspond to the class of midget or parvocellular (P) ganglion cells²⁷. A separate type of ganglion cell, displaying a center-only ON response to S-cone stimulation, possesses a unique bistratified morphology²⁸, with separate dendritic branches in the ON and OFF sublaminae of the inner plexiform layer. Discovery of this small bistratified ganglion cell and of its exclusively S-cone ON response provides strong evidence that the path taken from photoreceptor to ganglion cell by S-cone signals is separate from that taken by M- and L-cone signals.

Neuronal diversity and visual system channels

Retinal ganglion cells

The discovery of small bistratified cells and other ganglion cell types that innervate the LGN (Ref. 29) requires that a central dogma of primate visual system organization be modified. Most reviews of the primate visual system suggest that the retina sends two sorts of signals to the LGN (Fig. 2): a color-opponent signal arising from P cells, and a broad-band, luminance contrast signal from a population of larger parasol or magnocellular cells²⁷. These are seen as the beginning of parallel paths that make their way to the visual cortex³⁴. Yet a strictly bipartite division from ganglion cell to visual cortex is unlikely, not only, as many have shown, because the parallel channels converge somewhere in the cerebral cortex³⁵, but also because the assumption of an early dichotomy – of only two retinogeniculocortical channels – is in error. Instead, the discovery of several neurochemically distinct retinal circuits and of several retinogeniculate classes of ganglion cells²⁹ indicates that other channels exist in the monkey visual system.

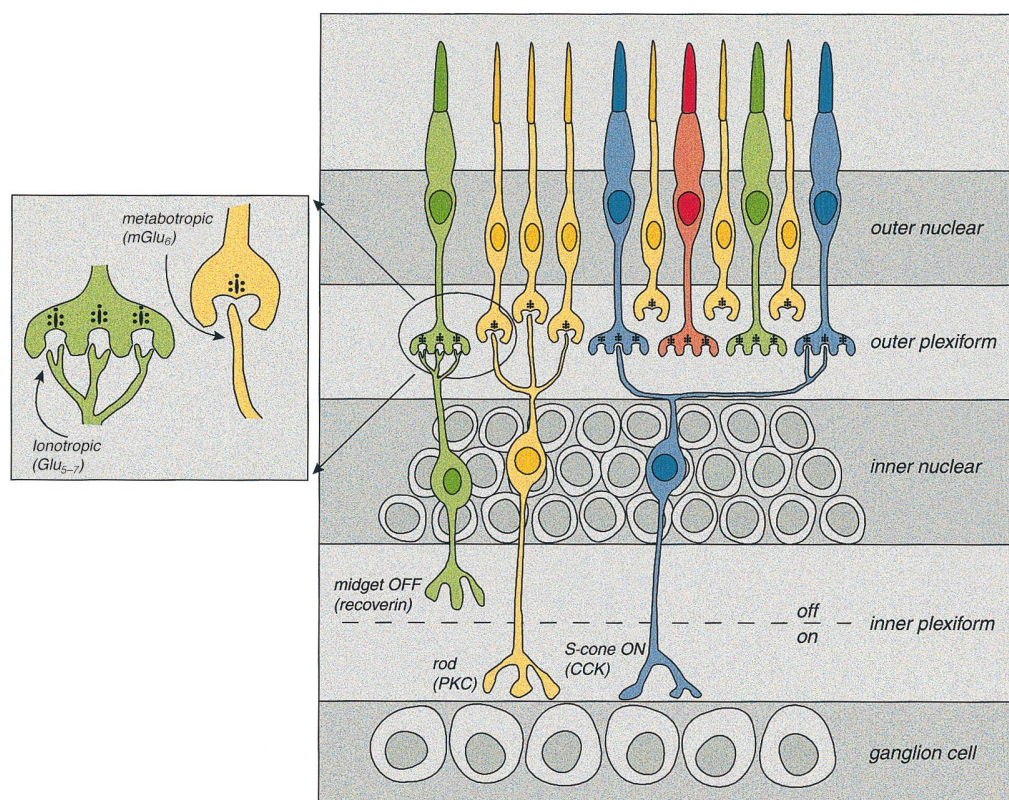


Fig. 1. Diagram of the hierarchical organization of the monkey retina showing three types of bipolar cell collecting from rods and three types of cone photoreceptor. The expression of the calcium-binding protein recoverin by midget OFF bipolar cells⁷, the enzyme protein kinase C (PKC) by rod bipolar cells⁹ and the neuroactive peptide cholecystinin (CCK) by S-cone ON bipolar cells⁸ distinguishes these cells from other types. (Inset) Sign-conserving responses to glutamate in some OFF bipolar cells is a result of their expression of ionotropic Glu₅₋₇ or kainate receptors, while sign-reversing responses in certain ON bipolar cells is a result of their expression of the metabotropic mGlu₆ or (±)-2-amino-4-phosphonobutyric acid (AP-4) receptor²³⁻²⁶.

Koniocellular (K) layers of the LGN

Studies of the primate LGN provide further evidence for a visual system that incorporates more than two channels, as patterns of molecular expression reveal the presence of at least three distinct geniculocortical populations occupying separate layers. Two of these populations are the neurons of the easily detected magnocellular and parvocellular layers, which differ from one another not only in morphological and physiological properties³² but also in patterns of immunocytochemical staining for a cell-surface protein³⁶. The third LGN population is composed of somata that are generally very small, thus deserving the name koniocellular (K), as applied to a homologous population in the LGN of prosimian bushbabies³⁷. A largely hidden population in macaques and other simians, K cells can be viewed in isolation because they express two proteins that cells of the magnocellular and parvocellular layers do not: the α subunit of type II calmodulin-dependent protein kinase (α CaM II kinase) and the 28 kDa vitamin D-dependent calcium-binding protein, calbindin^{38,39}. Studies exploiting this neurochemical signature have found that K layers occupy regions ventral to each of the magnocellular and parvocellular layers in macaques (Fig. 2), so that six of them exist through most of the LGN in this species³⁹.

What are the cells of the K layers doing? One answer is apparent in their cortical targets, as the K layers of the LGN provide the geniculocortical innervation of periodic patches in layer III of V1. These correspond to the cytochrome oxidase (CO)-rich puffs or

blobs^{37,39,40}. Furthermore, studies of macaques have found that LGN neurons retrogradely labeled by restricted injections of tracers into blobs express α CaM II kinase³⁹. Thus, combinations of anatomical tracing and immunocytochemical characterization demonstrate that K cells are the most direct route whereby retinal signals reach V1 blobs.

What those retinal signals relayed by macaque K cells might be is poorly understood. Some may be spectrally opponent responses much like the small bistratified ganglion cells of the retina^{32,33}, but some would be expected to carry mixed-cone responses. This incomplete understanding of K layers outside of bushbaby LGN arises both from a general failure to appreciate how robust the K layers are and from the difficulty of studying these layers without some objective means of identifying them. Both can be overcome by exploiting their selective expression of α CaM II kinase and calbindin.

Neurochemistry and function in visual cortex

Cell type-specific molecules?

Gamma-aminobutyric acid (GABA) and its synthesizing enzyme, glutamate decarboxylase, are molecules expressed by inhibitory interneurons in thalamus and cortex^{41,42}. If one defines function to include inhibition and excitation, then expression produces function for this type of neuron since release of GABA appears to be the one mechanism for producing synaptic inhibition in thalamus and cortex. Yet, when the functional properties of neurons in the LGN and visual cortex are usually discussed, they are said to include features such as luminance contrast sensitivity, motion detection and orientation selectivity. In this sense, inhibitory interneurons are identified with no single function since GABA neurons exist across M, P and K layers of macaque LGN (Ref. 42) and occupy all layers and compartments in areas of the visual cortex⁴¹. Even GABA subpopulations in visual cortex that display unique properties, such as parvalbumin immunoreactive chandelier cells⁴³, appear to contribute to many functions. Such findings support the conclusion that the functional unit of cerebral cortex is not the cell type but the dynamic group, composed of many types that participate in a common function. One might expect for cerebral cortex, then, that variations in molecular attributes would be seen across layers or columns or areas, as these represent the fundamental subdivisions of cortical organization and function. This expectation has been met at several levels.

Functional channels in visual cortex

Upon reaching the cerebral cortex, signals from M, P and K channels can be said to remain segregated or to become so thoroughly intermixed that the division

of labor in extrastriate areas bears little resemblance to the separate LGN signals that enter V1. For V1 and V2, the questions of circuitry and segregation of function hinge upon CO-rich blobs and interblobs in V1 and upon CO-rich thin and thick stripes and CO-poor pale stripes in V2 – compartments that were themselves recognized and studied because they are histochemically distinct⁴⁴. Some studies make the case for fundamental distinctions between blobs and interblobs in V1 (Refs 45,46) and among thick stripes, thin stripes and pale stripes in V2 (Ref. 47). Blobs in V1 and thin stripes in V2 are viewed as a continuation of a color system, interblobs in V1 and pale stripes in V2 as the formation of a system dealing with stimulus shape, and a magnocellular-dominated system, routed through layer IVB of V1 into thick stripes of V2, as part of a neural mechanism for motion processing⁴⁸. Other studies indicate, however, that the physiological differences between neurons in blobs and interblobs of V1 (Refs 49,50) and among neurons of thin stripes, thick stripes and pale stripes in V2 (Refs 51,52) are more limited. Arguments as to which view is closer to the truth have revolved around methods and strategies, particularly as to which visual stimuli are most appropriate for studying receptive fields of cortical neurons.

From a viewpoint provided by neuronal chemistry and synaptic circuits, CO-rich compartments in V1 and V2 scarcely resemble CO-poor compartments, as afferent terminations⁵³, intrinsic circuits⁵⁴ and efferent projections differ among them⁵⁵. Differences between blobs and interblobs extend to neurochemistry, as neurons expressing neuroactive substances such as glutamate and substance P, receptors such as GABA_A and glutamate receptor subunits, cytoplasmic proteins such as calbindin, parvalbumin and microtubule associated protein-2 and cell-surface proteins such as the Cat-301 protein are either enriched in or confined to neurons in blobs or interblobs⁵⁶. So only if neurons sample equally from the two compartments would reported similarities in blob and interblob physiological properties be consistent with the substantive differences in their circuits and molecules. Unfortunately, the data on this point are conflicting^{57,58}, and, until this fundamental anatomical issue is more persuasively resolved, an agnostic point of view concerning the physiological distinctions between CO-rich and CO-poor compartments appears advisable.

Extrastriate cortex

An overview of extrastriate cortex leaves a strong impression that functional precision arises out of anatomical homogeneity⁵⁹. Precision can be appreciated not only in the very different physiological properties displayed by neurons in what have been called dorsal and ventral streams^{59,60}, but also in the properties displayed by neighboring areas within a single stream^{61,62}. Yet extrastriate areas exhibit only minor variations in cell sizes or densities to mark the transition from one area to the next⁶⁰. To better document differences among areas, the best-case scenario would be the discovery of several molecules, each expressed by neurons in a single extrastriate area. More modest results have been reported. The cell-surface Cat-301 protein is expressed more richly in areas of the dorsal stream than of the ventral stream⁶³, and regional variations exist in the density or laminar distribution of neurons expressing calcium-binding proteins⁶⁴ or an epitope of the middle and high molecular weight

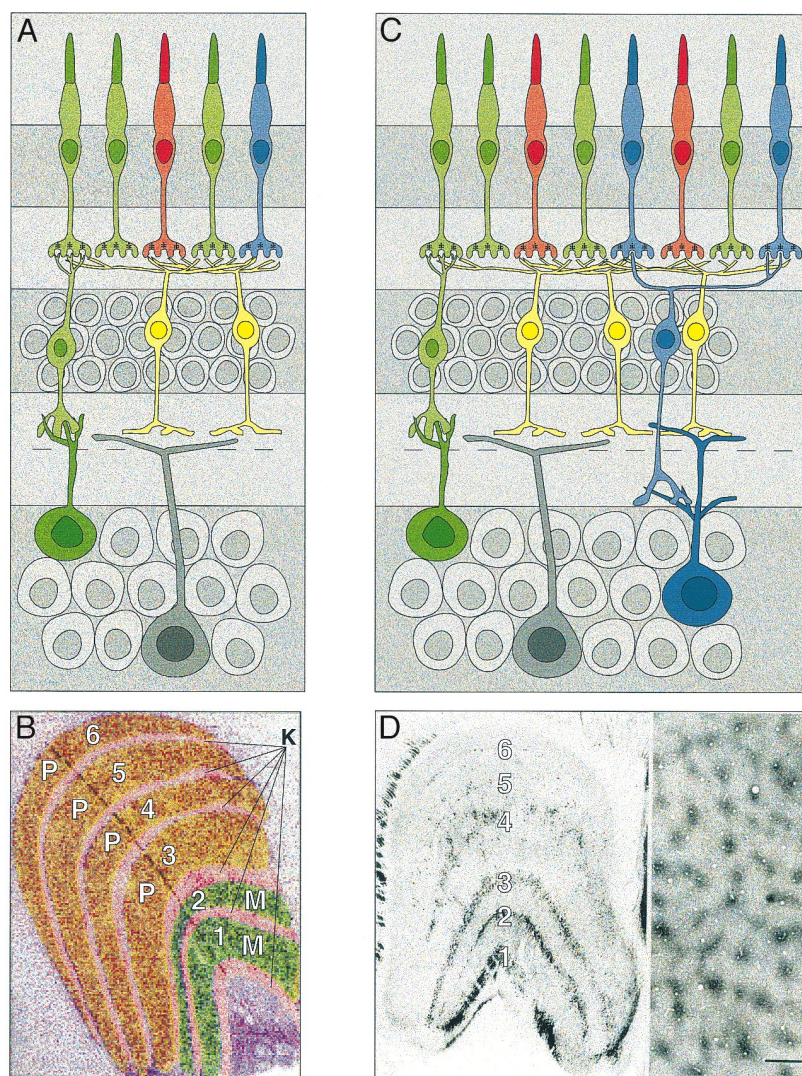


Fig. 2. Changing views of early visual pathways. (A) The classical view of the retina. Two simple 'midget' and 'diffuse' bipolar cell circuits provide input, respectively, to two ganglion cell types: midget (left) and parasol (right). (B) Midget and parasol ganglion cells project, respectively, to four parvocellular (P) and two magnocellular (M) layers of the lateral geniculate nucleus (LGN). Modified from Ref. 30. Conventional staining techniques did not reveal neurons in the koniocellular (K) layers ventral to each primary layer, thus strengthening the bipartite view of the early primate visual system. (C) The simple view of the retina is evolving to include the small bistratified ganglion cell that collects depolarizing (ON) signals from S cones and hyperpolarizing (OFF) signals from M and L cones^{28,31}. (D) Immunoreactivity for the α subunit of type II calmodulin-dependent protein kinase selectively marks the population of K neurons in macaque LGN (left). Some of these neurons probably receive retinal input in macaque from the small bistratified ganglion cells^{32,33}. K neurons provide the direct geniculocortical innervation of blobs in V1, histochemically marked by cytochrome oxidase activity (right). Scale bar, 200 μ m (left) and 350 μ m (right). Neurochemical characteristics permit the K layers and the blobs to be studied in isolation; each is a focal point for studies of parallel visual processing in monkeys.

neurofilament proteins⁶⁵. For most other proteins, peptides and amino acids, reported variations among extrastriate areas are minor^{66,67}.

That extrastriate areas express similar neurochemical patterns as well as a common cytoarchitecture can be interpreted in two ways. Either the prevailing view of extrastriate anatomy is correct – only in the afferents that provide their drive and in the targets that receive their messages do homotypical areas of cerebral cortex differ⁶⁸ – or the science itself is too limited. From the latter point of view, studies have been too few and have relied too heavily upon occasional good fortune to be of much predictive value; that is, the distinct possibility exists for robust differences among neurons of the many extrastriate visual areas, discovery

of which would accent the contribution of intrinsic differences^{69,70} to the physiology of extrastriate areas.

Neurochemical characterization and development

Rules that are evident for the visual system of adults apply to the visual system of fetuses and infants. Cell type-specific markers in retina permit strong statements to be made of when and how these types develop and the roles they play in the formation of retinal patterns and visual system organization. Two examples stand out. S cones and M or L cones express unique opsins very early in development, so at no time does there appear to be co-expression of these photopigments^{71,72}. Moreover, patterns of expression that are evident as the retina grows in concentric rings from center to periphery have led to the conclusion that the regular hexagonal array of photoreceptors is established by a precursor population of cones^{71,72}. Whether the 'organizer' population is the S cone⁷² or the M and L cones⁷¹ is open to debate. A more general statement about molecular expression and visual system organization has taken advantage of the ganglion cell type-specific expression of the *POU* gene products Brn 3a and Brn 3b (Ref. 73). That the adult pattern of immunostaining of these proteins is achieved in ganglion cells shortly after their last mitotic division⁷⁴ indicates a very early divergence in the formation of the M and P pathways in macaque retina. In both these examples it should be clear that the role played in retinal maturation by the respective proteins is unknown, and that these molecules are used here as a means to isolate neuronal types and to study them in isolation.

In contrast to type-specific expression during retinal development, it is expression in large neuronal populations in the development of the LGN and visual cortex that has been explored most thoroughly. Thus, studies of neurotransmitter receptors⁷⁵ and enzymes⁷⁶, of metabolic enzymes⁷⁷ and of calcium-binding proteins⁷⁸ have accentuated the establishment of an essential anatomical scaffold of areas, layers and compartments in fetal development, one that is modified with visual experience but that persists even in the absence of both retinae. Studies of neuronal proliferation have also accentuated the development of large neuronal ensembles, as gradients in the expression of gene products such as Pax-6 in the telencephalic neuroepithelium⁷⁹ have been interpreted as templates for development of distinct cortical areas.

Conclusions

With a perspective provided by neurochemical studies of retina, LGN, V1 and V2, and later extrastriate areas, something of a trend emerges. Circuits in retina incorporate unique and neurochemically distinct types of neurons to produce distinct functions. In LGN, V1 and V2 it is not individual types of neurons but groups laid out as distinct layers or compartments that continue to divide the labor of visual processing and to display unique molecular features. By the time most visual signals reach the later extrastriate areas they are processed by circuits that may be fundamentally similar across areas but are still likely to differ in molecular detail and in functional significance.

From a strategic point of view, the studies of neurochemical distinctions among types and groups of neurons in primate visual system offer something of an encouraging object lesson. Most studies have

relied upon luck and persistence to discover molecules expressed selectively by visual neurons. Nonetheless, the list of molecules and of the neurons they mark is impressive, and so is the increased understanding their discovery has provided of the organization and function of the primate visual system. Given the success of this near-random search program, a strong prediction is that a more systematic approach would be fruitful, particularly if it exploited modern molecular biological strategies for detecting differences in gene expression across populations of neurons. These strategies will most likely include means for constructing antibodies to antigens expressed by one group of animals and not another⁸⁰, of probing complementary DNA libraries constructed from very small, well-isolated groups of neurons and of targeting unique messenger RNAs by differential screening and subtractive methods^{81,82}.

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NO news from insect brains

Gerd Bicker

In nerve cells, the short-lived signalling molecule nitric oxide (NO) is generated by Ca²⁺–calmodulin-stimulated NO synthases. Nitric oxide activates soluble guanylate cyclase in target cells, leading to the formation of cGMP. Biochemical investigations have shown the presence of a Ca²⁺–calmodulin-regulated NO–cGMP signalling mechanism in the nervous system of insects. Using NADPH–diaphorase staining as a marker for the enzyme NO synthase and an antiserum against cGMP, the cellular organization of NO donor and target cells has so far been resolved in the locust and fruit fly. This paper provides an overview of the cellular organization of NO signalling in the insect nervous system as well as highlighting its functions in olfactory information processing, formation of olfactory memory, vision, and neuronal development. The resolution of discrete donor and NO-responsive target cells in the developing nervous system of *Drosophila* will facilitate the genetic and pharmacological analysis of NO–cGMP signal transduction.

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INTENSE RESEARCH over recent years has established that nitric oxide (NO), an unstable gas, serves as an endogenous messenger molecule in the nervous system and mediates a variety of unique functions as part of the processing of neural signals. Nitric oxide is an unconventional transmitter since it is not packaged in synaptic vesicles but, rather, diffuses from its site of production and moves readily through cell membranes. In nerve cells, NO is generated in an activity-dependent process by nitric oxide synthases (NOSs), Ca²⁺–calmodulin-stimulated enzymes^{1,2}. The principal function of NO appears to be as an activator

of soluble guanylate cyclase (sGC), one of the cGMP-synthesizing enzymes (Fig. 1A). The stimulation of sGC is caused by the binding of NO to the iron-containing haem group in the active centre of sGC. Despite the fact that NO diffuses in neuropilar compartments, the specificity of cellular communication is preserved by the activity-dependent release of the ligand and discrete distribution of the target receptor. Since NO is a free radical, other, cGMP-independent, signal-transduction pathways are also possible. Such an alternative perspective of NO signalling has recently been summarized by Stamler and colleagues⁴.

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